

AS Level Mathematics — Pure Mathematics 1 (P1)

Practice Paper 8

Time allowed: **1 hour 50 minutes**

Instructions to candidates

- Answer **all** questions. Write your answers on paper, showing all necessary working. Marks may be lost for insufficient working.
- If a numerical answer is not exact, give it correct to **3 significant figures**; give angles in degrees correct to **1 decimal place**, unless a different level of accuracy is specified in the question.
- A scientific calculator may be used. Do **not** use a graphical or symbolic calculator.
- The number of marks is given in brackets [] at the end of each question or part question. The total mark for this paper is **75**.
- This paper is written in the style of Cambridge International 9709 Paper 1 (Pure Mathematics 1). It is a practice paper, not an official examination paper.

List of formulae (as provided in the examination)

Quadratic equation: for $ax^2 + bx + c = 0$, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Binomial expansion: $(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + b^n$, where $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

Arithmetic series: $u_n = a + (n-1)d$, $S_n = \frac{1}{2}n(a + l) = \frac{1}{2}n\{2a + (n-1)d\}$

Geometric series: $u_n = ar^{n-1}$, $S_n = \frac{a(1-r^n)}{1-r}$ ($r \neq 1$), $S_\infty = \frac{a}{1-r}$ ($|r| < 1$)

Circular measure: arc length $= r\theta$, sector area $= \frac{1}{2}r^2\theta$ (θ in radians)

Trigonometry: $\tan \theta \equiv \frac{\sin \theta}{\cos \theta}$, $\sin^2 \theta + \cos^2 \theta \equiv 1$

Differentiation and integration: $\frac{d}{dx}(x^n) = nx^{n-1}$, $\int x^n dx = \frac{x^{n+1}}{n+1} + c$ ($n \neq -1$)

1 Solve the simultaneous equations

$$y = x^2 - 3x + 5, \quad y = 2x - 1.$$

[3]

2 Solve the equation $\cos\left(2x + \frac{\pi}{3}\right) = \frac{1}{2}$ for $0 \leq x \leq \pi$, giving your answers in terms of π . [4]

3 Find the coefficient of x^2 in the expansion of $(1+x)^4(2-x)^5$. [4]

4 The points A and B have coordinates $(1, -2)$ and $(5, 6)$ respectively.

(a) Find the equation of the perpendicular bisector of AB , giving your answer in the form $ax + by = c$. [2]

(b) The point C lies on the perpendicular bisector of AB and the area of triangle ABC is 20. Find the two possible positions of C . [3]

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(a) The sum of the first 20 terms of an arithmetic progression is 50 and the 20th term is -26 . Find the first term and the common difference. [3]

(b) A geometric progression has first term 3 and common ratio $-\frac{1}{2}$. Find the difference between the sum to infinity and the sum of the first 5 terms, giving your answer as an exact fraction. [3]

6 The function f is defined by $f(x) = \frac{3}{x+2}$ for $x > -2$.

(a) State the range of f . [2]

(b) Find an expression for $f^{-1}(x)$ and state the domain of f^{-1} . [2]

(c) The function g is defined by $g(x) = 2x - 1$ for $x \in \mathbb{R}$. Solve the equation $fg(x) = 1$. [2]

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(a) Given that $\tan \theta = \frac{3}{4}$ and $180^\circ < \theta < 270^\circ$, find the exact values of $\sin \theta$ and $\cos \theta$. [2]

(b) Prove the identity $\frac{1 - \sin \theta}{\cos \theta} \equiv \frac{\cos \theta}{1 + \sin \theta}$. [2]

(c) Hence solve the equation $\frac{1 - \sin \theta}{\cos \theta} = 2 \cos \theta$ for $0^\circ \leq \theta \leq 360^\circ$. [3]

8 A curve has equation $y = 4\sqrt{x} - x$ for $x > 0$.

(a) Find the coordinates of the stationary point of the curve and determine its nature. [3]

(b) Find the set of values of x for which y is an increasing function of x . [2]

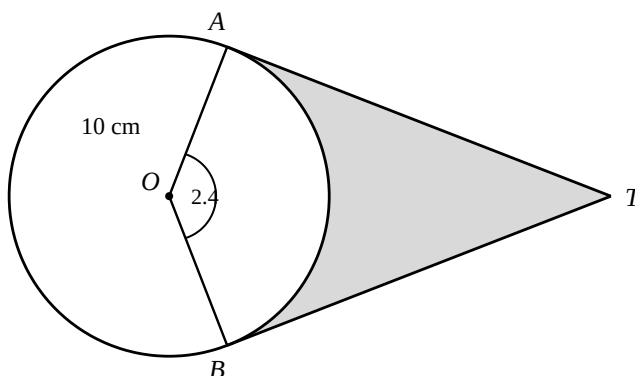
(c) A point is moving along the curve. At the instant when $x = 9$, the x -coordinate is increasing at a rate of 0.6 units per second. Find the rate of change of the y -coordinate at this instant. [2]

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(a) A curve is such that $\frac{dy}{dx} = (2x - 1)^2$ and the curve passes through the point $(1, 1)$. Find the equation of the curve. [4]

(b) Find the area of the region enclosed by the curve $y = 4 - x^2$ and the line $y = x + 2$. [4]

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The diagram shows a circle with centre O and radius 10 cm . The points A and B lie on the circle and angle $AOB = 2.4$ radians. The tangents to the circle at A and B meet at the point T . The region bounded by the tangents AT and BT and the minor arc AB is shaded.

- (a) Find the length of OT . [2]
- (b) Find the area of the shaded region. [3]
- (c) Find the perimeter of the shaded region. [3]

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- (a) Find the set of values of k for which the equation $x^2 + kx + k + 3 = 0$ has no real roots. [3]
- (b) Express $2x^2 - 4x + 5$ in the form $a(x + b)^2 + c$ and hence state the range of the function $f(x) = 2x^2 - 4x + 5$ for $x \in \mathbb{R}$. [3]
- (c) Find the values of the constant m for which the equation $2x^2 - 4x + 5 = mx$ has equal roots, giving your answers in exact form. [3]

12 A circle has centre $C(-2, 3)$ and touches the x -axis.

- (a) Find the equation of the circle. [2]
- (b) The line $y = 2x + 4$ meets the circle at the points P and Q . Find the coordinates of P and Q . [3]
- (c) Find the equation of the tangent to the circle at the point of intersection which is not on the x -axis, giving your answer in the form $ax + by = c$. [3]

END OF PAPER (Total: 75 marks)